Shyama Prasad Mukherji College for Women, University of Delhi
Department of Computer Science

Topic : Exploring Knowledge Boundaries of Large Language Models for Hallucination Reduction

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Introduction

- **Large Language Models (LLMs)** are widely used in today's time in various domains. It generates answers to a user's query based on its training data and the patterns it learnt. Such models are trained on large-scale data, and use it to generate responses.
- LLMs sometimes generate **plausible** but **factually incorrect answers**. This is called **Hallucinations**. Hallucination is an issue as the LLM may generate a factually wrong answer to a question whose answer it doesn't know. This may reduce **trust** and **reliability** on these models. Model may continue to create its own answer instead of accepting that it doesn't know.
- **Knowledge Boundary** is the range within which the model's knowledge/reliability is expected to be sufficient. A model can hallucinate even on information that is within its training knowledge. So, it becomes important to know about the knowledge boundary of models.

Motivation

Knowledge-boundary detection could work as a decision signal for when an LLM should rely on its **internal knowledge** and when it should **seek external information**.

Objective

The objective and possible research questions for this research can be :

1. if we **can identify** whether a question lies **within or outside** an **LLM's reliable knowledge boundary** ?
2. can this **information** be used to help **reduce hallucination**?

Literature Review

S.no	Year	Title	Citation Count	Key Contribution	Limitations
1.	2025	A Comprehensive Survey of Hallucination Mitigation Techniques in Large Language Models	7917	Introduces taxonomy for hallucination types and review causes, detection, benchmarks, RAG's limitations and open directions	• RAG Limitations are stated but not deeply analysed • Reproducibility is also difficult due to variety of dataset, models and metrics
2.	2024	Detecting hallucinations in large language models using semantic entropy	2580	Introduces **semantic entropy** with meaning-based clustering to detect hallucinations and improve detection (AUC 0.790) across 30 model-task settings	• Targets **Confabulations** where answers differ in meaning • Depends on multiple generations, extra computations semantic equivalence detection and task specific evaluation

Literature Review

S.no	Year	Title	Citation Count	Key Contribution	Limitations
3.	2022	Language Models (Mostly) Know What They Know	2228	• Introduces **P(IK)** to estimate if model knows the answer • Multiple samples improve **P(True)** but weaken on new tasks	• P(IK) reflect difficulty of question rather than true knowledge • Self Evaluation is inconsistent and fails on unseen tasks • RLHF may worsen miscalibration
4.	2023	Survey of Hallucination in Natural Language Generation	12509	Distinguishes **intrinsic and extrinsic hallucinations**, review causes and cross transfer opportunities	• Evaluation methods are difficult to compare directly • Limited detail on specific metrics and methods

Literature Review

S.no	Year	Title	Citation Count	Key Contribution	Limitations
5.	2026	Hallucination to Truth: A Review of Fact-Checking and Factuality Evaluation in Large Language Models	79	• Reviews factuality evaluation metrics and mitigation techniques • Highlights impact of dataset quality, and fine-tuning strategies on hallucinations	• Covers limited emerging approaches and rely on existing studies • Prevents direct comparison and identification of best approach under identical conditions
6.	2025	LLM Hallucinations in Practical Code Generation: Phenomena, Mechanism, and Mitigation	345	Examines code generation hallucinations and identifies contributing factors Proposes **RAG approach** for mitigation	• RAG is dependent on retrieved project information, validated only on specific LLMs

Literature Review

S.no	Year	Title	Citation Count	Key Contribution	Limitations
7.	2024	Discovering Latent Knowledge in Language Models Without Supervision	1039	• Introduces **Contrast-Consistent Search (CCS)** uses internal activations and logical consistency to assess truthfulness • It halves prompt sensitivity while maintaining accuracy under misleading prompts	• Covers limited emerging approaches and rely on existing studies • Prevents direct comparison and identification of best approach under identical conditions
8.	2023	Knowledge Injection to Counter Large Language Model (LLM) Hallucination	303	Introduces **Knowledge Injection (KI)** converts knowledge based info into LLM prompts, factual accuracy in responses.	• Dependent on reliable knowledge graphs and small evaluation less than 100 and covers narrow task-domain coverage

Literature Review

S.no	Year	Title	Citation Count	Key Contribution	Limitations
9.	2024	Large Language Models Must Be Taught to Know What They Don't Know	201	• Trains LLM on graded QA pairs for answer-confidence estimation outperforming baseline uncertainty methods with ~1,000 examples across 57 MMLU subjects.	• Limited to factual tasks, requires separate answer and uncertainty estimation with human decision-making evaluation
10.	2023	The Internal State of an LLM Knows When It's Lying	1339	• introduce **SAPLMA (Statement Accuracy Prediction, based on Language Model Activations)** • use LLM hidden-layer activations to detect truthfulness signals	• Limited to binary classification and only English-tested • Faces difficulty in handling long term responses.

Literature Review

S.no	Year	Title	Citation Count	Key Contribution	Limitations
11.	2024	A Survey of Confidence Estimation and Calibration in Large Language Models	503	Develops taxonomy for confidence-estimation methods and calibration metrics (Expected Calibration Error (ECE), AUROC) across text generation and classification	• Lacks experimental validation and evidence of effectiveness in multimodal LLMs.
12.	2020	Selective Question Answering under Domain Shift	371	Trains calibrator to predict incorrect answers in mixed question setting and addresses limitations of softmax-based confidence	• Limited to extractive QA, requires known out-of-the-domain data and achieve target accuracy on only 56% of questions

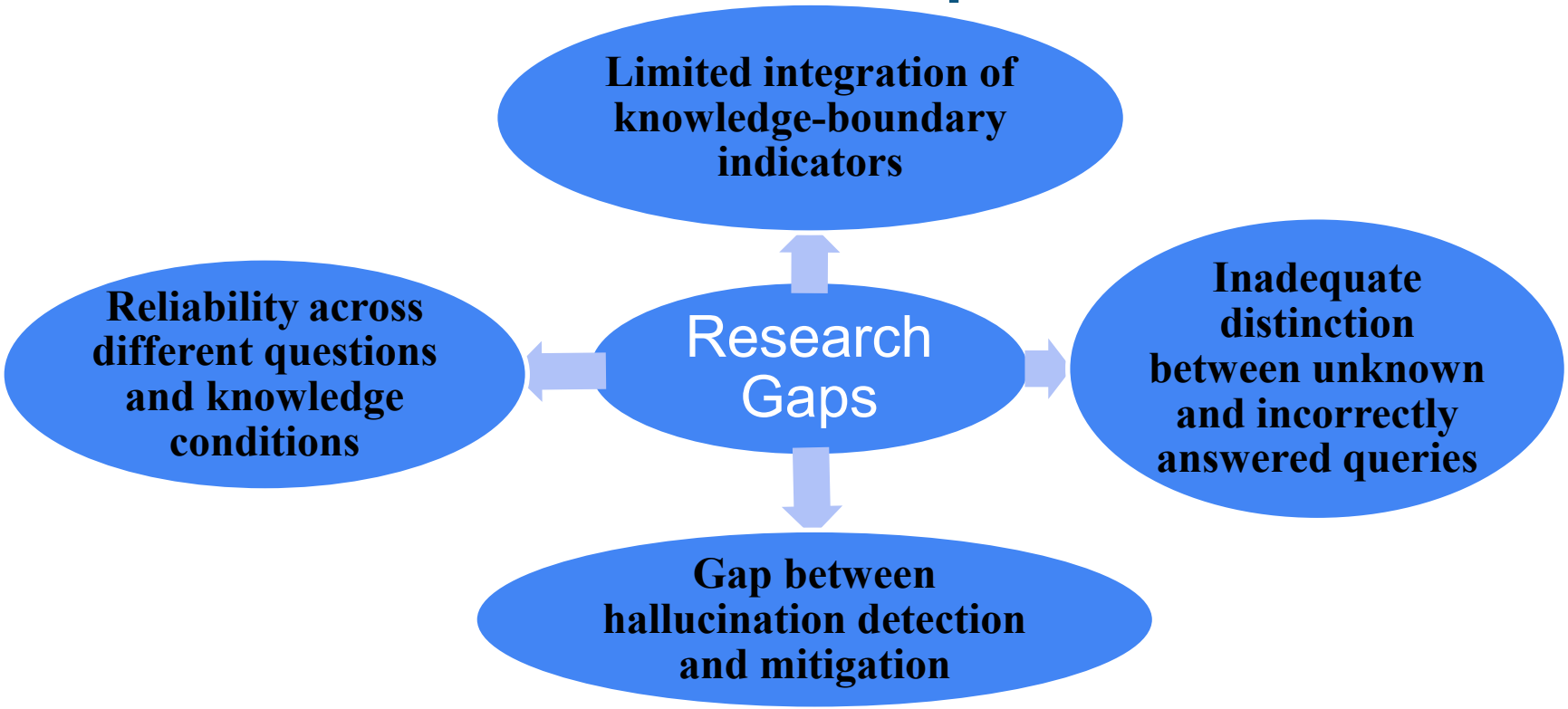
Literature Review

S.no	Year	Title	Citation Count	Key Contribution	Limitations
13.	2025	Benchmarking Knowledge Boundary for Large Language Models: A Different Perspective on Model Evaluation	54	• Distinguishes knowledge into prompt-agnostic, prompt-sensitive and unanswerable • Uses **Projected Gradient Descent with Constraints (PGDC)** to optimize question wording preserving its meaning	• Limited semantic preservation guarantees and cover only 5 models and small dataset
14.	2025	Teaching Large Language Models to Express Knowledge Boundary from Their Own Signals	37	Introduces **CoKE (Confidence-derived Knowledge boundary Expression)** 2 stages, using prior, direct and posterior awareness prompts to express LLM Knowledge Boundary	• Limited to short factual answers • Don't cover external knowledge or complex reasoning

Literature Review

S.no	Year	Title	Citation Count	Key Contribution	Limitations
15.	2025	Enhancing Uncertainty Modeling with Semantic Graph for Hallucination Detection	21	• Models uncertainty through semantic graphs, propagating it across entity relationships	• Limited to 2 datasets, dependent on many NLP components and lacks direct external fact verification

Research Gaps



Research Gaps

1. **Limited integration of knowledge-boundary indicators**

Existing works have explored and studied different signals to find if AI is uncertain. Our work further tries to investigate if these signals can work together to identify unreliable answers.

Example- The model gives the same answer five times, so semantic entropy is low. But the answer is wrong. Would another signal help us catch the error?

2. **Inadequate distinction between unknown and incorrectly answered queries**

Wrong answer by a model does not mean it lacks knowledge needed to answer. It can be due to knowledge limitation or answer-generation failure.

Example- An AI has encountered information about a historical event but gives the wrong year when asked about it.

Research Gaps

3. **Gap between hallucination detection and mitigation**

Identifying a potential wrong answer and actually preventing it are two different tasks.

Detection: “Answer may be unreliable”.

Mitigation: “Avoid presenting answer as certain, if it is not”

4. **Reliability across different questions and knowledge conditions**

A signal working well for one kind of question may not be working equally well for another kind.

Do the indicators work consistently across different types of factual questions, or do they fail in certain situations?

Research Questions

RQ1. To what extent can knowledge-boundary indicators, such as semantic entropy and confidence-related signals, identify whether an LLM can answer a factual question reliably?

RQ2. Can knowledge-boundary-aware answering decisions reduce hallucinations compared with standard LLM answering?

Methodology

1. **Dataset Selection**

- TriviaQA and Natural Questions(NQ-Open) datasets to be used for factual answering evaluation
- Select appropriate dataset splits and reference answers for assessing correctness

2. **Open-Source LLM selection**

- Lightweight open source LLMs like Ollama, will be selected according to available hardware.
- Generate answers to the selected questions under controlled prompting and inference settings.

Methodology

3. **Answer Correctness Evaluation**

- Compare generated answers with reference answers using exact match, token-level F1, and manual verification of ambiguous cases where required
- Categorize responses as correct, incorrect, or insufficiently supported.

4. **Knowledge-Boundary Estimation**

- Analyse confidence-related signals and semantic entropy to estimate answer uncertainty.
- Investigate whether these indicators distinguish reliable answers from incorrect or uncertain responses.

Methodology

5. **Hallucination Detection Analysis**

- Compare boundary indicators against actual answer correctness.
- Evaluate detection performance using precision, recall, F1-score, and AUROC where appropriate.

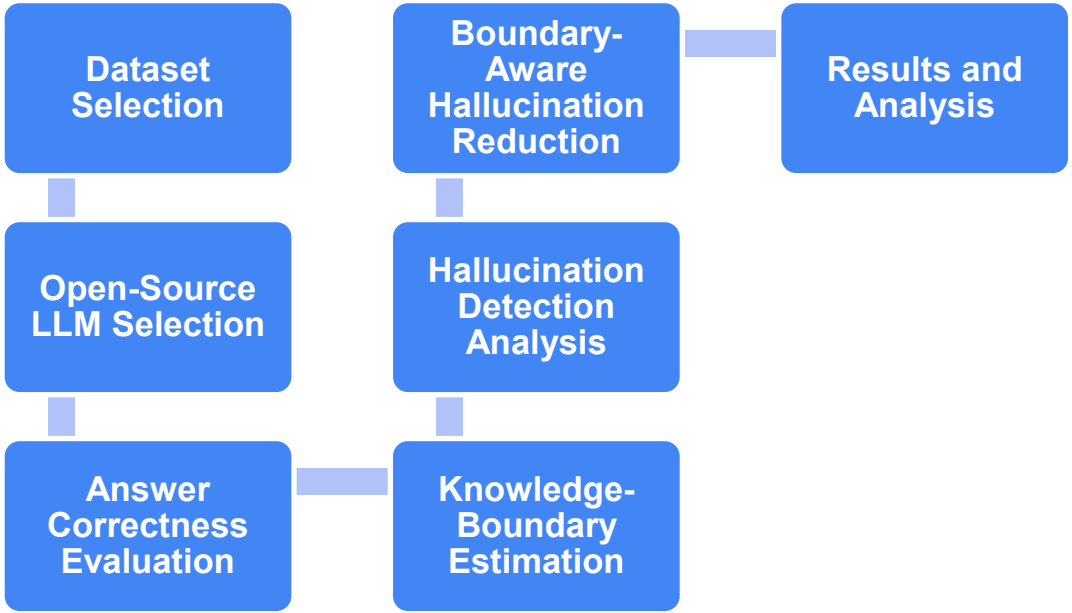
6. **Boundary-Aware Hallucination Reduction**

- Investigate a decision strategy that answers when reliability is sufficiently high and expresses uncertainty or seeks external evidence otherwise.
- Compare its performance with standard model answering using answer accuracy, hallucination rate, and answer coverage.

Methodology

7. **Results and Analysis**

- Analyse the effectiveness and limitations of the selected indicators.
- Examine whether knowledge-boundary estimation improves answer reliability and reduces hallucinations.



Techniques

Technique	Purpose
Semantic Entropy	Estimate uncertainty from semantic variation across answers
Confidence-related signals	Is model's confidence corresponding to answer's correctness?
Correctness-based evaluation	Does the model answers factual question accurately
Classification metrics	how effectively indicators identify unreliable answers
Threshold-based selective answering	Investigate when a model should answer, abstain, or seek external evidence

Further Work

1. **Experimental Setup:** We will be finalizing a LLM model and configure setup for experimental study
2. **Dataset Preparation:** Subsets and relevant attributes from TriviaQA and NQ-Open will be made for further experiments
3. **Model Evaluation:** Responses generated will be evaluated for correctness against reference answers.
4. **Knowledge-Boundary Estimation:** Implement and analyse semantic entropy and confidence-related indicators to estimate response reliability.
5. **Performance Evaluation:** Measure the effectiveness of boundary indicators using appropriate classification metrics.
6. **Hallucination Reduction:** Develop and evaluate a boundary-aware answering strategy against standard model responses.
7. **Results and Analysis:** Compare findings, identify limitations, and document the implications for reliable LLM question answering.

Conclusion

Large Language Models can sometimes give wrong answers confidently, which makes it difficult for users to know when to trust them. Existing techniques can help detect uncertainty and reduce hallucinations, but they cannot guarantee that every answer will be correct. Our research aims to find out whether we can identify when an LLM is likely to give a reliable answer and when its answer may be wrong. We will test this using open-source LLMs, factual question-answering datasets, and indicators such as semantic entropy and confidence. Finally, we will investigate whether using these indicators to guide the model's decisions can reduce incorrect answers and improve reliability.

References

1. [2401.01313] A Comprehensive Survey of Hallucination Mitigation Techniques in Large Language Models
2. Detecting hallucinations in large language models using semantic entropy | Nature
3. [2207.05221] Language Models (Mostly) Know What They Know
4. Survey of Hallucination in Natural Language Generation | ACM Computing Surveys
5. Hallucination to truth: a review of fact-checking and factuality evaluation in large language models | Artificial Intelligence Review | Springer Nature Link
6. LLM Hallucinations in Practical Code Generation: Phenomena, Mechanism, and Mitigation | Proceedings of the ACM on Software Engineering
7. Knowledge Injection to Counter Large Language Model (LLM) Hallucination | Springer Nature Link

References

8. [2212.03827] Discovering Latent Knowledge in Language Models Without Supervision
9. Large Language Models Must Be Taught to Know What They Don’t Know
10. The Internal State of an LLM Knows When It’s Lying - ACL Anthology
11. A Survey of Confidence Estimation and Calibration in Large Language Models - ACL Anthology
12. Selective Question Answering under Domain Shift - ACL Anthology
13. Benchmarking Knowledge Boundary for Large Language Models: A Different Perspective on Model Evaluation - ACL Anthology
14. Teaching Large Language Models to Express Knowledge Boundary from Their Own Signals - ACL Anthology

References

15. Enhancing Uncertainty Modeling with Semantic Graph for Hallucination Detection | Proceedings of the AAAI Conference on Artificial Intelligence
16. TriviaQA
17. GitHub - google-research-datasets/natural-questions: Natural Questions (NQ) contains real user questions issued to Google search, and answers found from Wikipedia by annotators. NQ is designed for the training and evaluation of automatic question answering systems. · GitHub
18. Google's Natural Questions